Aaron Lin

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EDUCATION

Boston University Boston, MA
B.S. Mechanical Engineering, Aerospace Concentration, Materials Science & Engineering Minor Sep 2024 – May 2028
GPA: 3.76/4.0, Dean's List
Completed Coursework: Mechanics of Materials, Materials Science, Energy and Thermodynamics, CAD and Machine Components, Probability and Statistics, Differential Equations, Computational Linear Algebra
Fall 2026: Fluid Mechanics, Dynamics, Electromechanical Design, Manufacturing Processes, Thermodynamics & Statistical Mechanics
Spring 2027: Aerodynamics, Measurement and Instrumentation, Structural Mechanics (Aerospace), Heat Transfer

PROFESSIONAL EXPERIENCE

Mechanical Engineering Intern Feb 2026 – Jun 2026
Culturon (plasma-activated coatings for life sciences) Sydney, Australia
- Designed and executed Optical Emission Spectroscopy (OES) experiments using an HR4000 spectrometer; authored repeatable protocols for plasma consistency evaluation and elemental contaminant detection.
- Developed an automated Python analysis suite (pandas, SciPy, Astroquery) to process OES spectral data, isolate elemental emission peaks, and cross-reference identifications against the NIST atomic line database.
- Authored a benchmarking report comparing OES plasma diagnostics against an established reference chamber and documenting key emission-line signatures.

AI Trainer Oct 2025 – Dec 2025
Handshake AI Fellowship Remote
- Developed and evaluated domain-specific prompts benchmarking LLM accuracy across engineering and physical science subfields; applied subject-matter expertise to identify reasoning failures on scientific and quantitative topics.

PROJECTS

Project Engineer Sep 2024 – Present
Boston University Rocket Propulsion Group Boston, MA
- Designing an aerodynamic fairing for an externally-mounted downcomer on the ARGO rocket; evaluating tangent-ogive and power-law nosecone profiles at a Mach 2.81 design condition using ANSYS Fluent CFD and OpenRocket trajectory simulation.
- Designed a converging-diverging rocket nozzle in SolidWorks using 1D compressible-flow theory; created part and assembly drawings and performed pressure-load FEA for an 80 psi chamber condition across three design reviews.
- Fabricated the nozzle via resin 3D printing and helped integrate it with a 1-inch NPT chamber, seals, relay ignition circuitry, and data acquisition for a team static fire.

CAD and Mechanical Layout Lead Sep 2026 – Oct 2026
2.5-DOF Cartesian Fishing Game Boston, MA
- Led SolidWorks CAD and mechanical layout for a joystick-controlled, belt-driven X–Y gantry with dual X rails, a Y carriage, and motorized Z-axis carrying an electromagnet.
- Designed and fabricated 3D-printed components to integrate stepper motors, idler pulleys, and linear carriages with 80/20 extrusions.
- Team demonstrated integrated X-Y-Z motion and magnetic pickup/release, for all three programmed game rounds.

Firmware and Integration Contributor Sep 2025 – Dec 2025
Autonomous Desk Cleaning Robot Boston, MA
- Authored Arduino C++ firmware implementing S-curve pathing, IR proximity-triggered 90° edge-avoidance turns, and gesture-activated start to enable autonomous operation across a 4 ft by 4 ft desk surface.
- Team ranked in the top 4 of approximately 20 groups; the documented final trial collected 72 of 95 rice grains with the upgraded fan and nozzle configuration.

SKILLS

Software: SolidWorks (Design, Simulation, HSMWorks), Fusion, Onshape, ANSYS (Fluent, Mechanical, Workbench), COMSOL, OpenRocket, Python, MATLAB, LaTeX, C++, Java
Engineering: Mechanical Design, FEA, CFD, CAD, OES, Supersonic Aerodynamics, 1D Compressible Flow, Stress Analysis, GD&T (fundamentals), Thermodynamic Cycle Analysis, Technical Writing, Systems Integration, Experimental Testing, Arduino
Fabrication: Additive Manufacturing (FDM, Resin), CNC Machining, Manual Lathe, Manual Mill, Soldering, Electronics Assembly
Languages: English (Native), Mandarin Chinese (Conversational)

CERTIFICATIONS

A Hands-On Introduction to Engineering Simulations — edX / CornellX, 2026
Machine Learning Specialization — DeepLearning.AI / Stanford University, 2024